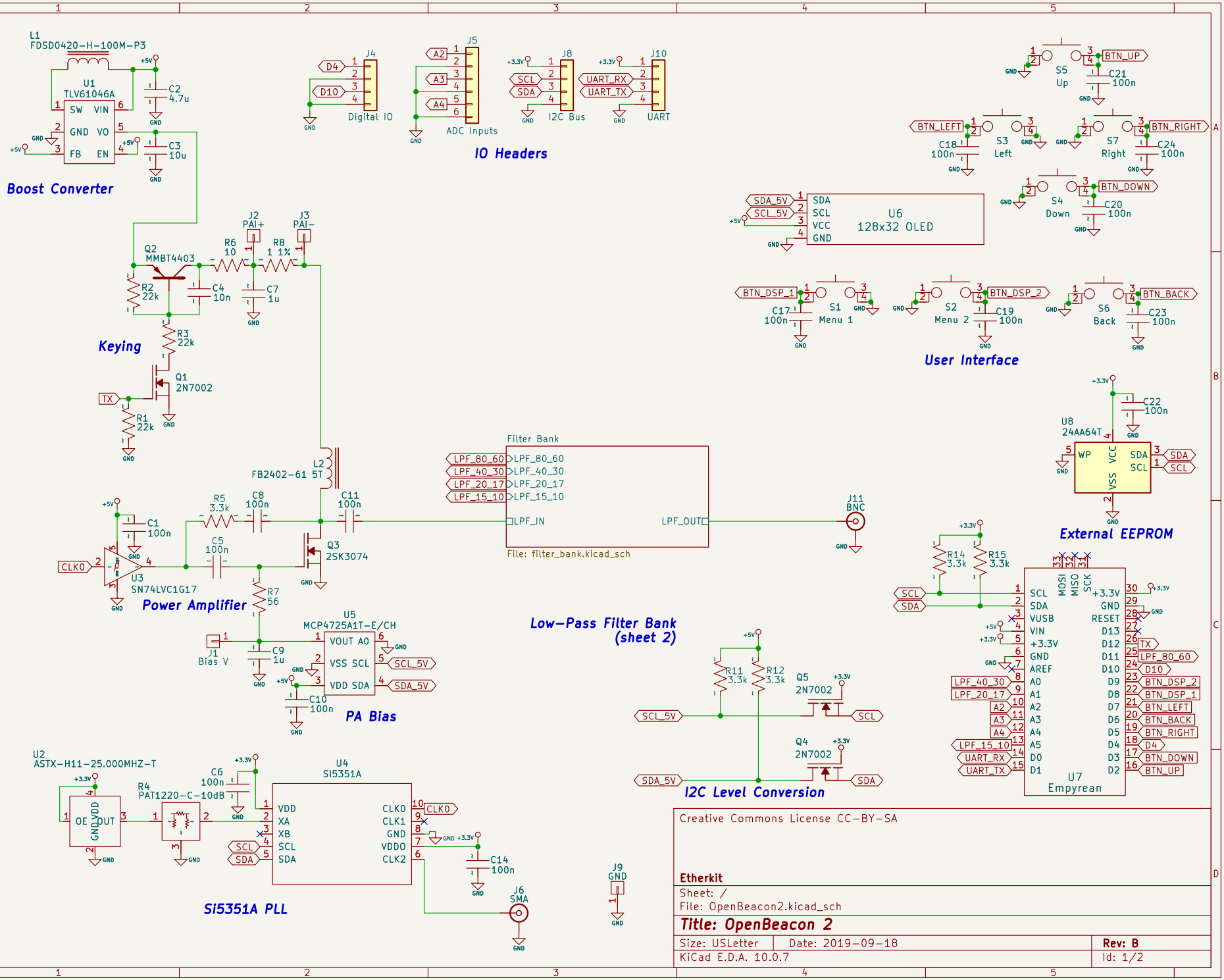
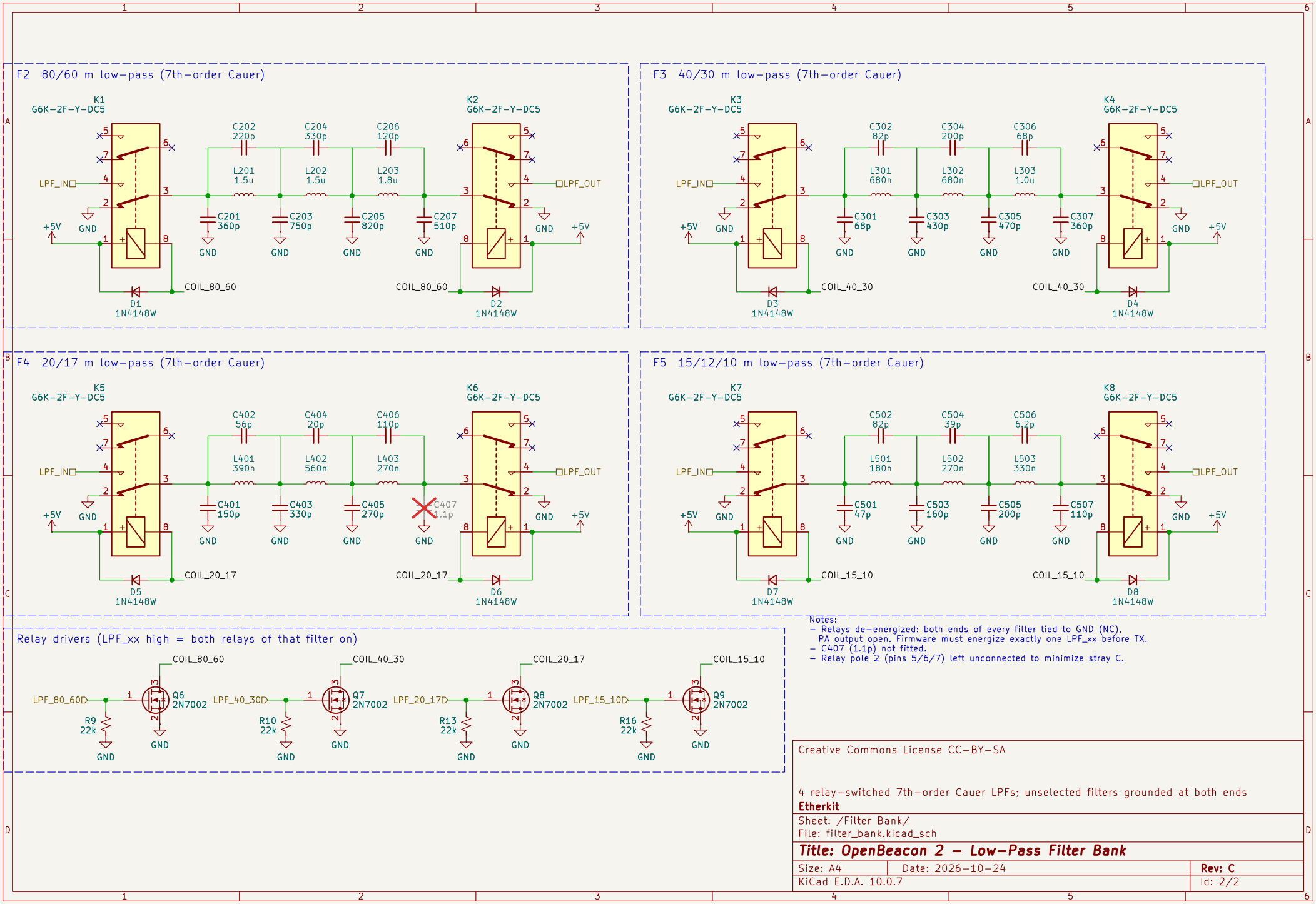

Boost Converter
IO Headers
Digital IO
ADC Inputs
I2C Bus
UART
User Interface
Keying
Power Amplifier
Filter Bank
File: filter_bank.kicad_sch
Low-Pass Filter Bank (sheet 2)
External EEPROM
PA Bias
I2C Level Conversion
SI5351A PLL
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Etherkit
Sheet: /
File: OpenBeacon2.kicad_sch
Title: OpenBeacon 2
Size: USLetter
Date: 2019-09-18
Rev: B
KiCad E.D.A. 10.0.7
Id: 1/2


F2 80/60 m low-pass (7th-order Cauer)
K1
G6K-2F-Y-DC5
K2
G6K-2F-Y-DC5
C202 220p
C204 330p
C206 120p
L201 1.5u
L202 1.5u
L203 1.8u
C201 360p
C203 750p
C205 820p
C207 510p
LPF_IN
LPF_OUT
+5V
GND
COIL_80_60
D1 1N4148W
D2 1N4148W
F3 40/30 m low-pass (7th-order Cauer)
K3
G6K-2F-Y-DC5
K4
G6K-2F-Y-DC5
C302 82p
C304 200p
C306 68p
L301 680n
L302 680n
L303 1.0u
C301 68p
C303 430p
C305 470p
C307 360p
COIL_40_30
D3 1N4148W
D4 1N4148W
F4 20/17 m low-pass (7th-order Cauer)
K5
G6K-2F-Y-DC5
K6
G6K-2F-Y-DC5
C402 56p
C404 20p
C406 110p
L401 390n
L402 560n
L403 270n
C401 150p
C403 330p
C405 270p
C407 1.1p
COIL_20_17
D5 1N4148W
D6 1N4148W
F5 15/12/10 m low-pass (7th-order Cauer)
K7
G6K-2F-Y-DC5
K8
G6K-2F-Y-DC5
C502 82p
C504 39p
C506 6.2p
L501 180n
L502 270n
L503 330n
C501 47p
C503 160p
C505 200p
C507 110p
COIL_15_10
D7 1N4148W
D8 1N4148W
Relay drivers (LPF_xx high = both relays of that filter on)
LPF_80_60
LPF_40_30
LPF_20_17
LPF_15_10
Q6 2N7002
Q7 2N7002
Q8 2N7002
Q9 2N7002
R9 22k
R10 22k
R13 22k
R16 22k
Notes:
- Relays de-energized: both ends of every filter tied to GND (NC).
PA output open. Firmware must energize exactly one LPF_xx before TX.
- C407 (1.1p) not fitted.
- Relay pole 2 (pins 5/6/7) left unconnected to minimize stray C.
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4 relay-switched 7th-order Cauer LPFs; unselected filters grounded at both ends
Etherkit
Sheet: /Filter Bank/
File: filter_bank.kicad_sch
Title: OpenBeacon 2 – Low-Pass Filter Bank
Size: A4
Date: 2026-10-24
Rev: C
KiCad E.D.A. 10.0.7
Id: 2/2